เลขที่นั่งสอบ

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 2 **Year**: 2018
Subject: 340152 Electrical Measuring Instruments **Section**: 5-6
Date: 1 March 2019 **Time**: 8.00-10.00
Name____________________ ID: ______________ Class________

Instructions:

1. The examination has 8 pages (including this page) and a total score of 70 points.
2. Write all your solution and answers on this examination sheet.
3. This is a **closed book** examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor' permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Q5) The following table of values represents a meter output in terms of the angular displacement of the needle, expressed in degrees, for a series of identical input currents. Determine the worst-case precision of the readings. (4 marks)

I_{in}(μA)	Output Displacement (Degrees)
10	20.10
10	20.00
10	20.20
10	19.80
10	19.70
10	20.00
10	20.30
10	20.10

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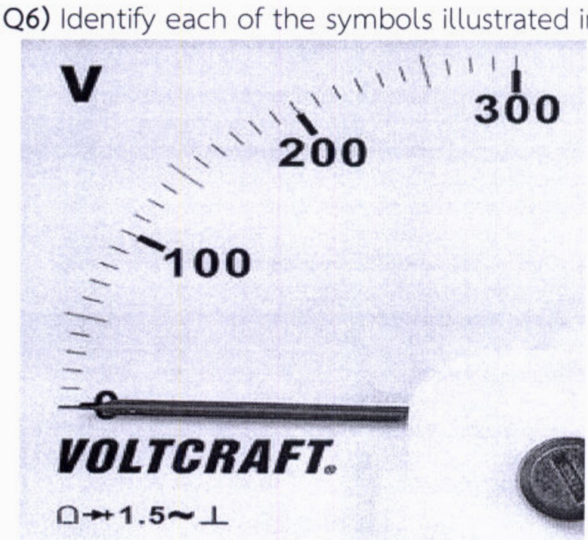
Q6) Identify each of the symbols illustrated in Figure. (3 marks)



a) ⊓ : ....

b) 1.5 : ....

c) ⊥ : ....

Q7) Define the following terms: (5 marks)

(a) Error : ....

(b) Accuracy : ....

(c) Precision : ....

(d) Measurement : ....

(e) Instrument : ....

Q8) State the three major categories of error. (2 marks)

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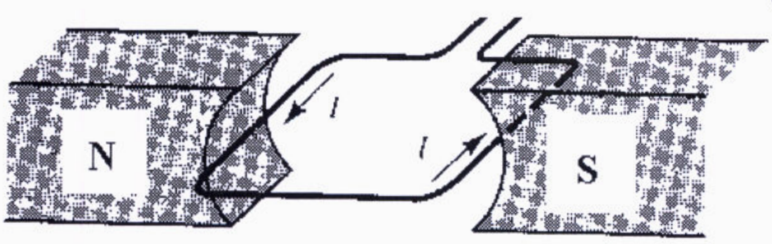
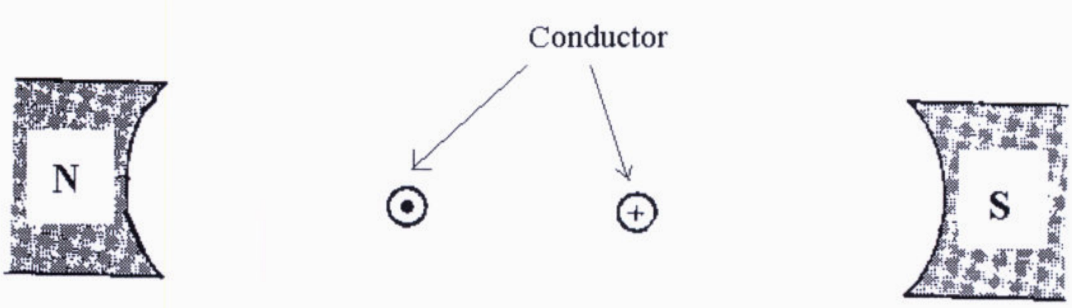
Q9) Using ***Right Hand Grip Rule*** to sketch the direction of the ***magnetic field*** and ***force***. (3 marks)





Q10) List and explain the **three forces** involving in the moving system of a deflection instrument. (3 marks)

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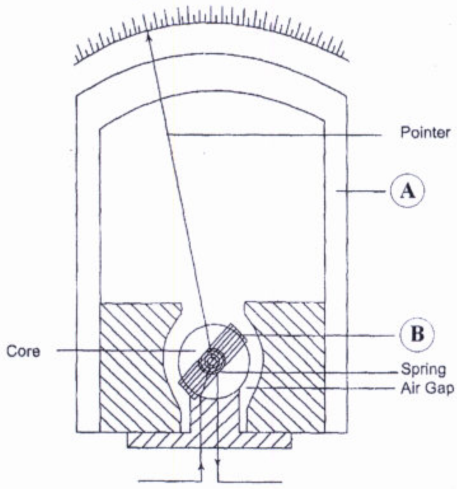
Q11) Describe the basic concept of meter movement. (3 marks)

Q12) The expected value of the voltage across a resistor is 50 V. However, the measurement gives a value of 49 V. Calculate (a) absolute error, (b) % error, (c) accuracy, and (d) % of accuracy. (5 marks)

Q13) Identify each part of the instrument: (2 marks)

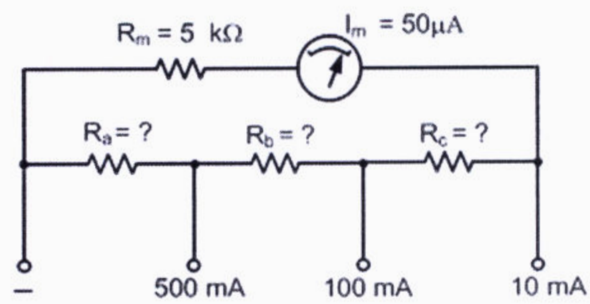


Part A is ....

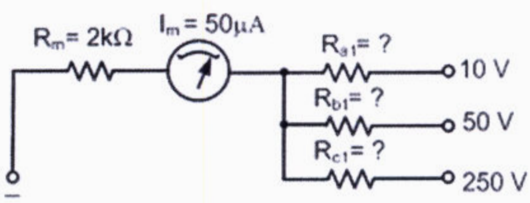
Part B is..

Assoc. Prof. Dr. Pichet Sriyanyong

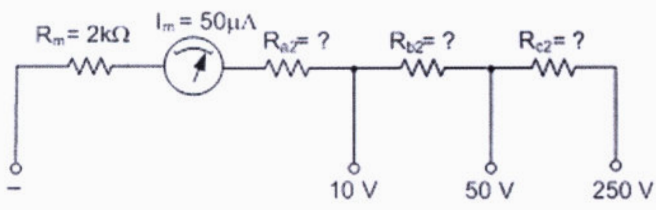
Q1) A PMMC instrument has FSD = 50 μA and R_m = 5 kΩ. Calculate the required resistance (R_a, R_b, R_c) values to convert the instrument into ***an ammeter*** (multi-range) with an FSD of 500 mA, 100 mA, and 10 mA (10 marks)



Q2) A PMMC instrument with FSD = 50 μA and R_m = 2 kΩ, is to be employed as ***a volt-meter*** with ranges of 10 V, 50 V and 250 V. ***Calculate*** the required values of multiplier resistors (R_a, R_b, R_c) for both cases and give the ***comparison*** to the practical volt-meter. (10 marks)



Case (a)



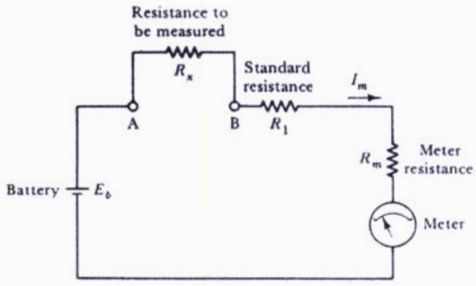
Case (b)

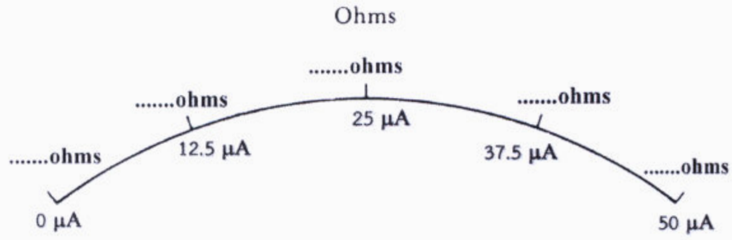
Q3) The series ohmmeter in figure is made up of a 1.5 V battery, a 50 μA meter, R_m = 2 kΩ.

(a) Calculate the value of R_1 to make the current to be full scale.

(b) Determine how the resistance scale should be marked at R_x = 0 and $R_x = \infty$.

(c) Determine how the resistance scale should be marked at 0.75 FSD, 0.5 FSD, and 0.25 FSD. (10 marks)





Q4) A voltmeter having a sensitivity of 1,500 Ω/V, reads 200 V on its 300 V scale when connected across an unknown resistor (Rx) in series with an ammeter and the ammeter reads 10 mA, Calculate the following items:

(a) The apparent resistance of the unknown resistor (R_T),

(b) The actual resistance of the unknown resistor (R_X),

(c) % error due to the loading effect. (10 marks)